

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet

range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

QUESTION 1: University Network Subnetting

Given:

- Network Address = 192.168.20.0/24
- Lab A = /26
- Lab B = /27
- Lab C = /28

Step 1: Lab A (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2 \\ &= 62 \text{ Hosts}\end{aligned}$$

Network Address : 192.168.20.0

First Host Address : 192.168.20.1

Last Host Address : 192.168.20.62

Broadcast Address : 192.168.20.63

Usable Hosts : 62

Step 2: Lab B (/27)

Subnet Mask = 255.255.255.224

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 27)} - 2 \\ &= 32 - 2 \\ &= 30 \text{ Hosts}\end{aligned}$$

Network Address : 192.168.20.64
 First Host Address : 192.168.20.65
 Last Host Address : 192.168.20.94
 Broadcast Address : 192.168.20.95
 Usable Hosts : 30

Step 3: Lab C (/28)

Subnet Mask = 255.255.255.240

Formula:

$$\begin{aligned}
 \text{Usable Hosts} &= 2^{(32 - 28)} - 2 \\
 &= 16 - 2 \\
 &= 14 \text{ Hosts}
 \end{aligned}$$

Network Address : 192.168.20.96
 First Host Address : 192.168.20.97
 Last Host Address : 192.168.20.110
 Broadcast Address : 192.168.20.111
 Usable Hosts : 14

Final Answer

Lab	Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Lab A	/26	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Interpretation of Results

The network 192.168.20.0/24 is successfully divided into three subnets using fixed subnet masks.

- Lab A (/26) provides 62 usable host addresses, making it suitable for a large laboratory.
- Lab B (/27) provides 30 usable host addresses, making it suitable for a medium-sized laboratory.
- Lab C (/28) provides 14 usable host addresses, making it suitable for a small laboratory.

Each subnet has a unique Network Address, Valid Host Range, and Broadcast Address, which improves IP address management, reduces network congestion, enhances security, and ensures efficient communication between devices.

QUESTION 2: Subnet Analysis for Two Systems

Given:

- System 1 IP Address = 192.168.1.130
- System 2 IP Address = 192.168.1.190
- Subnet Mask = 255.255.255.192 (/26)

Step 1: Determine the Subnet Size

Subnet Mask = /26

Number of Host Bits

$$= 32 - 26$$

$$= 6$$

Total IP Addresses per Subnet

$$= 2^6$$

$$= 64$$

Usable Host Addresses

$$= 64 - 2$$

= 62 Hosts

Subnet Increment

= 64

Therefore, the subnet ranges are:

- 192.168.1.0 – 192.168.1.63
- 192.168.1.64 – 192.168.1.127
- 192.168.1.128 – 192.168.1.191
- 192.168.1.192 – 192.168.1.255

Step 2: Analyze System 1

IP Address = 192.168.1.130

Network Address

= 192.168.1.128

Broadcast Address

= 192.168.1.191

Valid Host Range

= 192.168.1.129 – 192.168.1.190

Verification:

192.168.1.130 lies between
192.168.1.129 and 192.168.1.190.

Therefore,

System 1 belongs to the subnet

192.168.1.128/26.

Step 3: Analyze System 2

IP Address = 192.168.1.190

Network Address

= 192.168.1.128

Broadcast Address

= 192.168.1.191

Valid Host Range

= 192.168.1.129 – 192.168.1.190

Verification:

192.168.1.190 lies between

192.168.1.129 and 192.168.1.190.

Therefore,

System 2 also belongs to the subnet

192.168.1.128/26.

Final Result Table

System	IP Address	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	IP Within Subnet ?	Usable Hosts
System 1	192.168.1.130	255.255.255.192 (/26)	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62
System 2	192.168.1.190	255.255.255.192 (/26)	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62

Interpretation of Results:

Both System 1 (192.168.1.130) and System 2 (192.168.1.190) belong to the same subnet, 192.168.1.128/26. The valid host range for this subnet is 192.168.1.129 to 192.168.1.190, while the broadcast address is 192.168.1.191.

Since both IP addresses fall within the valid host range, they are valid host addresses and can communicate directly without requiring a router. Each /26 subnet provides 62 usable host addresses, making it suitable for medium-sized networks.

QUESTION 3: Organization Subnet Allocation

Given:

- Network Address = 172.16.0.0/16
- HR Department = /25
- Sales Department = /26
- IT Department = /27
- Admin Department = /28

Step 1: HR Department (/25)

Subnet Mask = 255.255.255.128

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 25)} - 2 \\ &= 128 - 2 \\ &= 126 \text{ Hosts}\end{aligned}$$

Network Address : 172.16.0.0

First Host Address : 172.16.0.1

Last Host Address : 172.16.0.126

Broadcast Address : 172.16.0.127

Usable Hosts : 126

Step 2: Sales Department (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2 \\ &= 62 \text{ Hosts}\end{aligned}$$

Network Address : 172.16.0.128

First Host Address : 172.16.0.129

Last Host Address : 172.16.0.190

Broadcast Address : 172.16.0.191

Usable Hosts : 62

Step 3: IT Department (/27)

Subnet Mask = 255.255.255.224

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 27)} - 2 \\ &= 32 - 2 \\ &= 30 \text{ Hosts}\end{aligned}$$

Network Address : 172.16.0.192

First Host Address : 172.16.0.193

Last Host Address : 172.16.0.222

Broadcast Address : 172.16.0.223

Usable Hosts : 30

Step 4: Admin Department (/28)

Subnet Mask = 255.255.255.240

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 28)} - 2 \\ &= 16 - 2 \\ &= 14 \text{ Hosts}\end{aligned}$$

Network Address : 172.16.0.224

First Host Address : 172.16.0.225

Last Host Address : 172.16.0.238

Broadcast Address : 172.16.0.239

Usable Hosts : 14

Remaining Unused IP Address Range

Start Address : 172.16.0.240

End Address : 172.16.255.255

Final Result Table

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR	/25 (255.255.255.128)	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26 (255.255.255.192)	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27 (255.255.255.224)	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28 (255.255.255.240)	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Interpretation of Results

The organization network 172.16.0.0/16 is successfully divided into four subnets based on departmental requirements.

- HR (/25) provides 126 usable host addresses.
- Sales (/26) provides 62 usable host addresses.

- IT (/27) provides 30 usable host addresses.
- Admin (/28) provides 14 usable host addresses.

The remaining IP address range (172.16.0.240 – 172.16.255.255) is available for future expansion. This subnetting approach improves IP address utilization, network performance, security, and simplifies network management.

QUESTION 4: Company Network Subnetting

Given:

- Network Address = 172.16.10.0/24
- Subnet Mask = /27 (255.255.255.224)
- IP Address 1 = 172.16.10.78
- IP Address 2 = 172.16.10.95

Step 1: Calculate the Number of Subnets

Original Prefix = /24

New Prefix = /27

Borrowed Bits

= $27 - 24$

= 3 bits

Number of Subnets

= 2^3

= 8 Subnets

Step 2: Calculate the Number of Usable Host

Host Bits

$$= 32 - 27$$

$$= 5 \text{ bits}$$

Usable Hosts

$$= 2^5 - 2$$

$$= 32 - 2$$

$$= 30 \text{ Hosts}$$

Step 3: Determine the Subnet of IP Address 172.16.10.78

Subnet Increment

$$= 32$$

Subnet Ranges

172.16.10.0 – 172.16.10.31

172.16.10.32 – 172.16.10.63

172.16.10.64 – 172.16.10.95

172.16.10.96 – 172.16.10.127

172.16.10.128 – 172.16.10.159

172.16.10.160 – 172.16.10.191

172.16.10.192 – 172.16.10.223

172.16.10.224 – 172.16.10.255

IP Address = 172.16.10.78

Therefore,

Network Address

$$= 172.16.10.64$$

Valid Host Range

= 172.16.10.65 – 172.16.10.94

Broadcast Address

= 172.16.10.95

Step 4: Check Whether 172.16.10.95 Belongs to the Same Subnet

IP Address = 172.16.10.95

172.16.10.95 is the Broadcast Address of the subnet

172.16.10.64/27.

Therefore,

It belongs to the same subnet but cannot be assigned to a host because it is reserved as the broadcast address.

Final Result Table

Description	Result
Network Address	172.16.10.0/24
Subnet Mask	/27 (255.255.255.224)
Number of Subnets Created	8
Usable Hosts per Subnet	30
IP Address Analyzed	172.16.10.78
Subnet (Network Address)	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
Does 172.16.10.95 belong to the same subnet?	Yes
Is 172.16.10.95 a valid host address?	No (It is the Broadcast Address)

Interpretation of Results

The company network 172.16.10.0/24 is subnetted using a /27 subnet mask, creating 8 equal-sized subnets with 30 usable host addresses each.

The IP address 172.16.10.78 belongs to the subnet 172.16.10.64/27. Its valid host range is from 172.16.10.65 to 172.16.10.94, and its broadcast address is 172.16.10.95.

Although 172.16.10.95 belongs to the same subnet, it is reserved as the broadcast address and cannot be assigned to any device. This subnetting scheme improves network organization, security, and efficient IP address utilization.

QUESTION 5: IPv6 Addressing

Given:

- IPv6 Address = 2001:0db8:0000:0000:0000:ff00:0042:8329
- Prefix Length = /64

Step 1: Compress the IPv6 Address

Original Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Compression Rules:

- Remove leading zeros in each block.
- Replace consecutive groups of all zeros with "::" (only once).

Compressed Address

2001:db8::ff00:42:8329

Step 2: Expand the IPv6 Address

Compressed Address

2001:db8::ff00:42:8329

Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix

Prefix Length = /64

The first 64 bits represent the Network Prefix.

Network Prefix

2001:db8:0:0::/64

Step 4: Calculate the Number of Host Addresses

Host Bits

= 128 – 64

= 64

Number of Possible Host Addresses

= 2^{64}

= 18,446,744,073,709,551,616

Final Result Table

Description	Result
Original IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed IPv6 Address	2001:db8::ff00:42:8329
Expanded IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:db8:0:0::/64
Host Bits	64
Total Possible Host Addresses	18,446,744,073,709,551,616 (2^{64})

Interpretation of Results

The given IPv6 address is successfully compressed by removing leading zeros and replacing consecutive zero blocks with "::". Expanding the compressed address restores the original full notation.

Using a /64 prefix, the first 64 bits identify the network, while the remaining 64 bits are used for host addressing. This provides 2^{64} (18,446,744,073,709,551,616) possible host addresses, allowing IPv6 to support a very large number of devices with efficient routing and scalable network communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation